

Chapter 3

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Exercise 1.

Let $\phi : Y \rightarrow X$ be a nonconstant holomorphic map of Riemann surfaces, with X connected. Show that for all $x \in X$ we have

$$\sum_{y \in \phi^{-1}(x)} e_y = n,$$

where n is the cardinality of the fibers not containing branch points, and e_y is the ramification index at $y \in Y$.

Proof.

WLOG assume that Y is connected as well, as we can sum over each connected component.

If x does not lie below branch points of Y , then we are immediately done as $\sum_{y \in \phi^{-1}(x)} e_y = \sum_{[Y:X]} 1 = n$.

Now suppose that x has a branch point of Y . For each point y_i in the fiber of x , there exists a neighbourhood V_i and a complex chart $g_i : V_i \rightarrow \mathbb{C}, f_i : \phi(V_i) \rightarrow \mathbb{C}$ such that $(f_i \circ \phi \circ g_i^{-1})(z) = z^{e_{y_i}}$ when expressed in this chart. Hence, pick a x_0 that is in a sufficiently small (punctured) open neighbourhood of x such that all its fibers land in V_i . Then:

$$\begin{aligned} n = \#[\phi^{-1}(x_0)] &= \sum_i \#[\phi^{-1}(x_0) \cap V_i] \\ &= \sum_i \#[(f_i \circ \phi \circ g_i^{-1})(f_i(x_0))] \\ &= \sum_i \#[(z \mapsto z^{e_{y_i}})^{-1}(f_i(x_0))] \\ &= \sum_i e_{y_i} \\ &= \sum_{y \in \phi^{-1}(x)} e_y. \quad \blacksquare \end{aligned}$$

Exercise 2.

Consider a nonconstant holomorphic map $\phi : P^1(\mathbb{C}) \rightarrow P^1(\mathbb{C})$.

(a) Show that there exists a unique rational function $f \in \mathbb{C}(t)$ such that $\phi = \phi_f$ as in the proof of Proposition 3.3.11.

Proof.

For existence, by Corollary 3.3.12. there exists an anti-isomorphism $\text{Hom}(P^1(\mathbb{C}), P^1(\mathbb{C})) \cong \text{Hom}(\mathcal{M}(P^1(\mathbb{C})), \mathcal{M}(P^1(\mathbb{C})))$ that sends $\phi \mapsto \phi^*$. Hence we are reduced to understanding $\mathbb{C}$ -linear embeddings $\phi^* : \mathbb{C}(t) \hookrightarrow \mathbb{C}(t)$. But then this is entirely determined by the image of t ; which is necessarily some rational function f . Hence $\phi(z) = \phi^*(z) = f(z)$.

For uniqueness, suppose that f, g are two such rational functions. Then $\frac{f}{g}$ is a rational function that is 1 everywhere except for the poles of f, g . By continuity $\frac{f}{g} = 1$ everywhere, so $f = g$. ■

(b) Relate the branch points of ϕ_f to zeros and poles of f , and determine the ramification indices.

Proof.

First suppose that $z_0 \in \mathbb{C} \setminus P(f)$. Pick a sufficiently small neighbourhood U of z_0 such that neither U , $V := f(U)$ contains infinity. We apply the appropriate linear transformation to U and V to center both at the origin:

$$\begin{array}{ccc} U & \xrightarrow{f} & V \\ \downarrow z - z_0 & & \downarrow z - f(z_0) \\ U_0 & \dashrightarrow & V_0 \end{array} \qquad \begin{array}{ccc} z + z_0 & \longmapsto & f(z + z_0) \\ \uparrow & & \downarrow \\ z & \dashrightarrow & f(z + z_0) - f(z_0) \end{array}$$

then z_0 is a branch point if and only if the zero of $f(z + z_0) - f(z_0)$ at the origin is not simple; i.e. $\lim_{z \rightarrow 0} \frac{f(z + z_0) - f(z_0)}{z} = f'(z_0) = 0$. The ramification index then is exactly $1 + \text{ord}_{z_0}(f')$.

Equally, when $z_0 \in P(f) \cup \{\infty\}$ we can apply the same logic but instead with the function $f(\frac{1}{z})$, to get the following result: A point $p \in P^1(\mathbb{C})$ is a branch point if and only if it is a zero or pole of $f'(z)$, with ramification index exactly equal to one plus its order as a pole or zero of $f'(z)$ ■

(c) Show that ϕ_f is a holomorphic automorphism of $P^1(\mathbb{C})$ if and only if $f = (at + b)(ct + d)^{-1}$ with some $a, b, c, d \in \mathbb{C}$ satisfying $ad - bc \neq 0$. [Hint: Observe that if ϕ_f is an automorphism, f can only have a single zero and pole, and these must be of order 1.]

Proof.

Let f be such an automorphism. Then it can necessarily only have a single zero and pole, which must be of order 1. Hence we can write $f(t) = \frac{at+b}{ct+d} = \frac{a}{c} \frac{t+b/a}{t+d/c}$. This function is nonconstant iff the pole and the zero do not cancel each other out; which is true iff $\frac{b}{a} \neq \frac{d}{c}$ iff $ad - bc \neq 0$.

Finally, we verify that such nonconstant f always define automorphisms of $P^1(\mathbb{C})$: For surjectivity, note that the image of f must be open by the Open Mapping Theorem. The image must also be compact, but as $P^1(\mathbb{C})$ is Hausdorff it must be closed. By the connectedness of $P^1(\mathbb{C})$ we are done. Injectivity follows from the fact that the (formal) inverse of f given by $f^{-1}(t) = \frac{dt-b}{-ct+a}$ can take on at most one value. ■

(d) Deduce that a holomorphic automorphism of $P^1(\mathbb{C})$ has one or two fixed points.

Proof.

By the previous result, let $f = \frac{at+b}{ct+d}$ be a holomorphic automorphism of $P^1(\mathbb{C})$. Then $f(t) = t$ reduces to a quadratic in t , such that f has one or two fixed points (depending on the root multiplicity of the resultant quadratic). ■

Exercise 3.

Let $Y \rightarrow X$ be a holomorphic map of compact Riemann surfaces, restricting to a cover $Y' \rightarrow X'$ outside the branch points.

(a) Show that the étale $\mathcal{M}(X)$ -algebra $\mathcal{M}(Y)$ is isomorphic to a finite direct sum of copies of $\mathcal{M}(X)$ if and only if the cover $Y' \rightarrow X'$ is trivial.

Proof.

The given statement is not true if you do not assume X is connected: Take $X = P^1(\mathbb{C}) \sqcup P^1(\mathbb{C})$, and $Y = P^1(\mathbb{C}) \sqcup (P^1(\mathbb{C}) \sqcup P^1(\mathbb{C}))$. This is a trivial cover with $\mathcal{M}(X) \cong \mathbb{C}(t) \oplus \mathbb{C}(t)$ and $\mathcal{M}(Y) \cong \mathbb{C}(t) \oplus \mathbb{C}(t) \oplus \mathbb{C}(t)$. One has $\mathbb{C}$ -dimension 2 and the other has 3, so $\mathcal{M}(Y)$ cannot be a free module over $\mathcal{M}(X)$.

Thus, for the rest of this question (including (b)) we will be assuming that X is connected.

As $\mathcal{M}(Y) \cong \prod \mathcal{M}(Y_i)$, WLOG assume that Y is connected such that we are reduced to the case of considering the extension $\mathcal{M}(Y) = \mathcal{M}(X)$. Note that by compactness there must be finitely many connected components. By the equivalence of categories in Theorem 3.3.7., the extension is trivial iff Y' is degree 1 cover of X' , i.e. is X' itself. Hence $\mathcal{M}(Y)$ decomposes into a finite direct sum iff each connected component of $Y' \rightarrow X'$ is trivial, i.e. the cover is trivial. ■

We make the additional observation that the number of copies of $\mathcal{M}(X)$ is exactly equal to the degree of the covering. Hence the question statement can be strengthened to the following:

The étale $\mathcal{M}(X)$ -algebra $\mathcal{M}(Y)$ is isomorphic to a direct sum of n copies of $\mathcal{M}(X)$ if and only if the cover $Y' \rightarrow X'$ is trivial of degree n .

This will be relevant in (b) below.

(b) Using Chapter 2, Exercise 2 give another proof of the fact that in the anti-equivalence of Theorem 3.3.7 Galois branched covers correspond to Galois extensions.

Proof.

We start with some finite extension $\mathcal{M}(Y)/\mathcal{M}(X)$. By the primitive element theorem, write $\mathcal{M}(Y) = \mathcal{M}(X)[\alpha]$ for some α . In the case where the extension is Galois, denote the Galois group by G .

First notice that this extension is Galois if and only if the minimal polynomial m_α splits completely in $\mathcal{M}(Y)$, as $\mathcal{M}(X)[\alpha] \cong \mathcal{M}(X)[t]/(m_\alpha(t))$ would be the splitting field of m_α .

Next we show that $\mathcal{M}(Y) \otimes_{\mathcal{M}(X)} \mathcal{M}(Y) \cong \bigoplus_{|G|} \mathcal{M}(Y)$ if and only if $\mathcal{M}(Y)/\mathcal{M}(X)$ is Galois: We have that $\mathcal{M}(Y) \otimes_{\mathcal{M}(X)} \mathcal{M}(Y) \cong \mathcal{M}(X)[t]/(m_\alpha) \otimes_{\mathcal{M}(X)} \mathcal{M}(Y) \cong \mathcal{M}(Y)[t]/(m_\alpha)$ as $\mathcal{M}(X)$ -algebras. By the Chinese remainder theorem, $\mathcal{M}(Y)[t]/(m_\alpha)$ splits into a finite direct sum of quotients by the irreducible factors of m_α . Hence each component is isomorphic to $\mathcal{M}(X)$ if and only if the factor is linear; i.e. m_α splits completely. In particular, when this does happen, m_α must split into exactly $\deg m_\alpha = [\mathcal{M}(Y) : \mathcal{M}(X)] = |G|$ many factors.

Next we show that $\mathcal{M}(Y \times_X Y) = \mathcal{M}(Y) \otimes_{\mathcal{M}(X)} \mathcal{M}(Y)$. By Theorem 3.3.7. we have that finite Galois branched covers are anti-equivalent to finite Galois extensions of $\mathcal{M}(X)$ via the $\mathcal{M}(-)$ functor. Notice that the fibered product over X is exactly the product in the slice category of spaces over X , and the tensor product is the coproduct in the category of rings/modules. But then $\mathcal{M}$ is a (contravariant) equivalence, so it commutes with the product to form the coproduct.

By the result of (a), $(Y \times_X Y)'$ is isomorphic to a degree $|G|$ trivial cover if and only if $\mathcal{M}(Y \times_X Y) \cong \bigoplus_{|G|} \mathcal{M}(Y)$.

Next notice that $(Y \times_X Y)' = Y' \times_{X'} Y'$, viewing both as subsets of $Y \times Y$. This step is straightforward, as the cover $(Y \times_X Y)' \rightarrow X'$ is obtained via removing all the points that shares fibers with a branch point.

By the result of Chapter 2 Exercise 2, $Y' \rightarrow X'$ is H -Galois if and only if $Y' \times_{X'} Y' \cong Y' \times H$ over Y' .

We have now proven every statement necessary for the question. To spell it out:

$$\begin{aligned}
 \mathcal{M}(Y)/\mathcal{M}(X) \text{ Galois} &\iff \mathcal{M}(Y) \otimes_{\mathcal{M}(X)} \mathcal{M}(Y) \cong \bigoplus_{|G|} \mathcal{M}(Y) \\
 &\iff \mathcal{M}(Y \times_X Y) \cong \bigoplus_{|G|} \mathcal{M}(Y) \\
 &\iff (Y \times_X Y)' \text{ is isomorphic to a degree } |G| \text{ trivial cover} \\
 &\iff Y' \times_{X'} Y' \cong Y' \times G \\
 &\iff Y' \times_{X'} Y' \cong Y' \times H \\
 &\iff Y' \rightarrow X' \text{ is } H\text{-Galois}
 \end{aligned}$$

where $H := \text{Aut}(Y'|X')$, of which we know that $|G| = |H|$ via Theorem 3.3.7. ■

Exercise 4.

Let X be a compact connected Riemann surface, and $\phi : X \rightarrow P^1(\mathbb{C})$ a holomorphic map. Show that if ϕ is not an isomorphism, then its branch points cannot all lie above the same point of $P^1(\mathbb{C})$. [*Hint:* Use the Riemann–Hurwitz formula.]

Proof.

By the Riemann-Hurwitz formula,

$$2g_X - 2 = d(2g_Y - 2) + \sum_y (e_y - 1)$$

where the sum is taken over all the branch points of ϕ . Now suppose that $Y = P^1(\mathbb{C})$ and all the branch points lie above some common point x_0 . Then

$$2g_X - 2 = -2d + \sum_{y \in \phi^{-1}(x_0)} e_y - \#\phi^{-1}(x_0)$$

$$2g_X - 2 = -2d + d - \#\phi^{-1}(x_0)$$

where $\sum_{y \in \phi^{-1}(x_0)} e_y = d$ by the result of Exercise 1. But then $2g_X - 2 = -d - \#\phi^{-1}(x_0) < -1 - 1 = -2$, such that $g_X < 0$. But then this is not possible. ■

Exercise 5.

Let $n > 2$ be an even integer, and consider the dihedral group

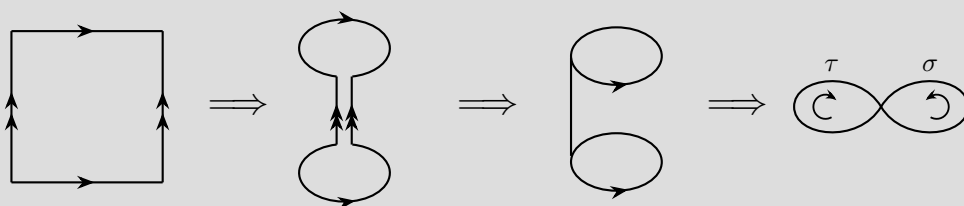
$$D_n = \langle \sigma, \tau \mid \sigma^n = \tau^2 = 1, \sigma\tau = \tau\sigma^{n-1} \rangle$$

Show that every complex torus $X = \mathbb{C}/\Lambda$ has a Galois branched cover $Y \rightarrow X$ with group D_n with exactly n branch points, all lying above the same point of X .

Proof.

Fix some parallelogonal fundamental domain for the lattice Λ , and start by removing some point p in the interior of the parallelogram.

Then the punctured parallelogram is homotopic to the empty square, which we see to be homeomorphic to the figure eight as follows:



The fundamental group of the figure eight is $\langle \tau \rangle * \langle \sigma \rangle \cong \mathbb{Z} * \mathbb{Z}$. Hence we get a quotient $\pi_1(X, x) \rightarrow D_n$ (where $x \neq p$ is some arbitrary point) with $\tau \mapsto \tau, \sigma \mapsto \sigma$. By Theorem 2.3.7. this corresponds to a cover $Y' \rightarrow X \setminus \{p\}$, which is Galois with automorphism group D_n by Theorem 2.2.10.. By Proposition 3.2.6. this cover naturally extends to a Galois branched cover of compact Riemann surfaces $\phi : Y \rightarrow X$. It remains to determine the number of branch points.

By Proposition 3.2.10., all the points above p are branch points of the same ramification index e , and by the result of Exercise 1 we see that $\deg \phi = e \cdot \#\phi^{-1}(p)$ so it is sufficient to determine e .

For each branch point y lying above p , pick a local chart $(V, g) \xrightarrow{\phi} (U, f)$ such that ϕ is locally of the form $z \mapsto z^e$. By Example 2.4.12., we know that $\pi_1(U, p)$ is generated by a simple loop γ around p ; and the monodromy action of γ is trivial if and only if it is a multiple of e . Hence e is exactly the smallest positive integer such that γ^e has the trivial action.

To that end, by following the homeomorphism depicted in the diagram, a counterclockwise loop around the square corresponds to the loop $\sigma\tau\sigma^{-1}\tau^{-1}$, which is equivalent to σ^{-2} under the quotient. Hence we are equipped to prove a stronger statement:

$$e = \text{ord}(\sigma^{-2}) = \begin{cases} m & n = 2m \\ n & n = 2m + 1 \end{cases}$$

and thus

$$\#\phi^{-1}(p) = \frac{2n}{e} = \begin{cases} 4 & n = 2m \\ 2 & n = 2m + 1 \end{cases}. \quad \blacksquare$$

Exercise 6.

Let $\phi : \mathbb{C}/\Lambda \rightarrow \mathbb{C}/\Lambda'$ be a holomorphic map of complex tori.

(a) Show that ϕ has no branch points.

Proof.

Note that complex tori have genus equal to 1. Hence by the Riemann Hurwitz formula, $2 - 2 = d(2 - 2) + \sum_y (e_y - 1)$ and $\sum_y (e_y - 1) = 0$. This is only possible when there are no branch points. ■

(b) Denoting by $p : \mathbb{C} \rightarrow \mathbb{C}/\Lambda$, $p' : \mathbb{C} \rightarrow \mathbb{C}/\Lambda'$ the natural projections, show that there exists a holomorphic map $\tilde{\phi} : \mathbb{C} \rightarrow \mathbb{C}$ with $p' \circ \tilde{\phi} = \phi \circ p$.

Proof.

For visibility we denote $\tilde{\phi}$ instead by Φ .

Fix some $z_0 \in \mathbb{C}$. If such a Φ exists, we must necessarily have that $p'(\Phi(z_0)) = \phi(p(z_0))$, such that $\Phi(z_0)$ must be in the fiber of p' above $\phi(p(z_0))$. Fix some $w_0 \in p'^{-1}(\phi(p(z_0)))$.

$$\begin{array}{ccc} \mathbb{C} & \xrightarrow{\quad \Phi \quad} & \mathbb{C} \\ p \downarrow & & \downarrow p' \\ \mathbb{C}/\Lambda & \xrightarrow[\quad \phi \quad]{} & \mathbb{C}/\Lambda' \end{array}$$

As $\mathbb{C}$ is simply connected, it is the universal cover of $\mathbb{C}/\Lambda$ by Proposition 2.4.9.. Hence given w_0 , by the lifting property of the universal cover there exists a unique lift Φ with $\Phi(z_0) = w_0$ that makes the above diagram commute.

To see that Φ is holomorphic, fix some $z_0 \in \mathbb{C}$. Then let U, V denote sufficiently small neighbourhoods of $p(z)$ and $\phi(p(z))$ such that both covers p, p' are trivialized. Let U_0, V_0 denote the unique sheet above U, V containing z_0 and $\Phi(z_0)$. Then this gives us an atlas on which Φ is locally homeomorphic to a holomorphic function. Thus Φ is itself necessarily holomorphic. ■

(c) Check that setting $\tilde{\phi}(\infty) = \infty$ defines an extension of $\tilde{\phi}$ to a holomorphic map $P^1(\mathbb{C}) \rightarrow P^1(\mathbb{C})$, and show that this map is in fact a holomorphic automorphism of $P^1(\mathbb{C})$. [Hint: Use Exercise 4.]

Proof.

To show that Φ extends holomorphically in the manner described, it suffices to show that there exists $k \geq 0$ such that $\lim_{z \rightarrow \infty} \frac{1}{z^k} \Phi(z) = 0$ exists; i.e. Φ is meromorphic at infinity. Take $k = 2$.

To that end, fix a basis $\{\omega_1, \omega_2\}$ and a fundamental domain D for Λ . Then every complex number z can be uniquely written of the form $z_0 + n\omega_1 + m\omega_2$ for $z_0 \in D$ and n, m integers. Now consider the function $\Phi(z) - \Phi(z + \omega_i)$ ($i = 1, 2$). As $p' \circ \Phi = \phi \circ p$, by chasing the diagram in (b) we see that $p' \circ \Phi$ is a constant function. But then by the open mapping theorem and the discreteness of Λ' this is only possible when $\Phi(z) - \Phi(z + \omega_i) = c_i$ is constant. Putting everything together, we see that

$$\frac{1}{z^k} \Phi(z) = \frac{\Phi(z_0) + nc_1 + mc_2}{(z_0 + n\omega_1 + m\omega_2)^k}.$$

For z to tend to infinity, one of or both of $|n|, |m|$ must go to infinity as well. In the case where $|n| \rightarrow \infty$ only, we see that

$$\lim_{z \rightarrow \infty} \left| \frac{1}{z^2} \Phi(z) \right| = \lim_{|n| \rightarrow \infty} \frac{|\Phi(z_0)/n^2 + c_1/n + mc_2/n^2|}{|z_0/n + \omega_1 + m\omega_2/n|^2} \rightarrow 0.$$

The proof for the other two cases is analogous. So Φ indeed extends holomorphically to $P^1(\mathbb{C})$.

By the result of (a), this extended function can only have branch points above infinity. Hence by the result of Exercise 4, ϕ is necessarily an isomorphism. ■

(d) Establish a correspondence between holomorphic maps $\phi : \mathbb{C}/\Lambda \rightarrow \mathbb{C}/\Lambda'$ and pairs $(a, b) \in \mathbb{C}^2$ with $a \neq 0$ and $a\Lambda + b \subset \Lambda'$. [Hint: Use Exercise 2.]

Proof.

The statement of the question is incorrect. The correspondence should be between pairs $(a, b) \in \mathbb{C}^2$ with $a \neq 0$ and $a\Lambda \subseteq \Lambda'$. As we will see there is no condition on b .

By the result of Exercise 2(c), and part (c) above, isomorphisms $P^1(\mathbb{C}) \rightarrow P^1(\mathbb{C})$ are described exactly by (nondegenerate) Möbius transformations.

First, suppose we have some $\phi : \mathbb{C}/\Lambda \rightarrow \mathbb{C}/\Lambda'$. Write $\Phi(z) = \frac{az+b}{cz+d}$. As $\Phi(\infty) = \infty$ we have that $c = 0$, so Φ must be in fact an affine function of the form $az + b$ where $a \neq 0$. In particular, as $p' \circ \Phi = \phi \circ p$, we get that $p'(\Phi(z)) = p'(\Phi(z + \omega))$ for all $\omega \in \Lambda$; i.e. $\Phi(z) - \Phi(z + \omega) \in \Lambda'$. By substituting in $z = 0$ we get that $a\Lambda \subseteq \Lambda'$.

It remains to verify that such (a, b) -pairs always induce well-defined maps between the complex tori. To that end, it is sufficient to show that $\Phi(z) = az + b$ lifts through the quotients; i.e. for all $\omega \in \Lambda$, $\Phi(z) - \Phi(z + \omega) \in \Lambda'$. But then if $a\Lambda \subseteq \Lambda'$, we have that $\Phi(z) - \Phi(z + \omega) = -a\omega \in \Lambda'$. ■